SMOIS SESI 2022/2023

SECTION A

① a) $\lim\limits_{x \to -3} \dfrac{x^3+27}{x+3}$

	$x^2 - 3x + 9$
$x+3$	$x^3 + 0x^2 + 0x + 27$
$(-)$	$x^3 + 3x^2$
	$-3x^2 + 0x$
$(-)$	$-3x^2 - 9x$
	$9x + 27$
	$9x + 27$
	$\cdot \quad \cdot$

K1 long division/ factorize

$= \lim\limits_{x \to -3} \dfrac{(x+3)(x^2-3x+9)}{x+3}$ J1

$= \lim\limits_{x \to -3} x^2 - 3x + 9$

$= (-3)^2 - 3(-3) + 9$ K1

$= 27$ J1

b) $\lim\limits_{x \to 2} \dfrac{\sqrt{x^2+12} - 4}{x-2}$

$= \lim\limits_{x \to 2} \dfrac{\sqrt{x^2+12} - 4}{x-2} \times \dfrac{\sqrt{x^2+12} + 4}{\sqrt{x^2+12} + 4}$ K1 conjugate

$= \lim\limits_{x \to 2} \dfrac{(x^2+12) - 16}{(x-2)(\sqrt{x^2+12} + 4)}$

$= \lim\limits_{x \to 2} \dfrac{x^2 - 4}{(x-2)(\sqrt{x^2+12} + 4)}$ J1 simplify numerator

$= \lim\limits_{x \to 2} \dfrac{(x-2)(x+2)}{(x-2)(\sqrt{x^2+12} + 4)}$ K1 factorize denominator

$= \lim\limits_{x \to 2} \dfrac{x+2}{\sqrt{x^2+12} + 4}$

$= \dfrac{2+2}{\sqrt{2^2+12} + 4}$ K1 sub value

$= \dfrac{1}{2}$ J1

Total : 9m

② $$\frac{1}{(x-1)^2(x-3)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-3}$$ B1

$$1 = A(x-1)(x-3) + B(x-3) + C(x-1)^2$$ K1 ✓

when $x=1$
$1 = B(1-3)$
$B = -\frac{1}{2}$

when $x=3$
$1 = C(3-1)^2$
$C = \frac{1}{4}$

when $x=0$,
$1 = A(-1)(-3) - \frac{1}{2}(-3) + \frac{1}{4}(-1)^2$
$A = -\frac{1}{4}$

K1 ✓ at least one correct

$$\therefore \frac{1}{(x-1)^2(x-3)} = -\frac{1}{4(x-1)} - \frac{1}{2(x-1)^2} + \frac{1}{4(x-3)}$$ J1

$$y = -\frac{1}{4(x-1)} - \frac{1}{2(x-1)^2} + \frac{1}{4(x-3)}$$

$$y = -\frac{1}{4}(x-1)^{-1} - \frac{1}{2}(x-1)^{-2} + \frac{1}{4}(x-3)^{-1}$$

$$\frac{dy}{dx} = \frac{1}{4}(x-1)^{-2} + (x-1)^{-3} - \frac{1}{4}(x-3)^{-2}$$ K1 ✓

$$\frac{dy}{dx} = \frac{1}{4(x-1)^2} + \frac{1}{(x-1)^3} - \frac{1}{4(x-3)^2}$$ J1

$$\frac{d^2y}{dx^2} = -\frac{1}{2}(x-1)^{-3} - 3(x-1)^{-4} + \frac{1}{2}(x-3)^{-3}$$ K1 ✓

$$\frac{d^2y}{dx^2} = -\frac{1}{2(x-1)^3} - \frac{3}{(x-1)^4} + \frac{1}{2(x-3)^3}$$ J1

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = -\frac{1}{2(-1-1)^3} - \frac{3}{(-1-1)^4} + \frac{1}{2(-1-3)^3}$$

$$= -\frac{17}{128}$$ J1

Total : 9m

③ $f(x) = x^4 - 8x^3 + 18x^2 - 27$

$f'(x) = 4x^3 - 24x^2 + 36x$ K1

$f'(x) = 0$

$4x^3 - 24x^2 + 36x = 0$ K1

$4x(x^2 - 6x + 9) = 0$

$4x(x-3)^2 = 0$

$4x = 0 \qquad (x-3)^2 = 0$

$x = 0 \qquad x = 3$

critical numbers are $x=0$ and $x=3$ J1

$f''(x) = 12x^2 - 48x + 36$ K1

when $x = 0$,

$f''(0) = 36 > 0$ (minimum point)

when $x = 3$,

$f''(3) = 12(3)^2 - 48(3) + 36 = 0$

test for inflection point

interval	$(-\infty, 3)$	$(3, \infty)$
value x	2	4
sign $f''(x)$	$-$	$+$
conclusion	concave downward	concave upward

K1 any test

$\therefore$ There is inflection point at $x=3$.

$f(3) = 3^4 - 8(3)^3 + 18(3)^2 - 27 = 0$ K1

$\therefore$ Point of inflection at $(3, 0)$. J1

Total: 7m

SECTION B

1 a) LHS: $\overline{z_1 - z_2} = \overline{(6+3i)-(5-2i)}$

$= \overline{1+5i}$

$= 1-5i$ K1

RHS: $\overline{z_1} - \overline{z_2} = (6-3i)-(5+2i)$ K1

$= 1-5i$

$\therefore \overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$ shown J1

b) $z_1 = 6+3i$

$\arg z_1,\ \theta = \tan^{-1}\left|\frac{3}{6}\right|$ K1

$= 0.4363$ rad J1

Total : 5m

2 a) $x^{(\ln x-2)} = e^3$

$\ln x^{(\ln x-2)} = \ln e^3$ K1 ln both sides

$(\ln x-2)(\ln x) = 3$ B1

$(\ln x)^2 - 2\ln x - 3 = 0$ K1 J1

let $u = \ln x$

$u^2 - 2u - 3 = 0$

$(u-3)(u+1) = 0$

$u=3, \quad u=-1$

$\ln x = 3, \quad \ln x = -1$ K1

$x = e^3 \quad x = e^{-1}$ J1 J1

b) $|2x+3| > |x-5|$, $x \neq 5$

$$\left(\sqrt{(2x+3)^2}\right)^2 > \left(\sqrt{(x-5)^2}\right)^2$$

$$(2x+3)^2 > (x-5)^2$$

$$4x^2+12x+9 > x^2-10x+25$$

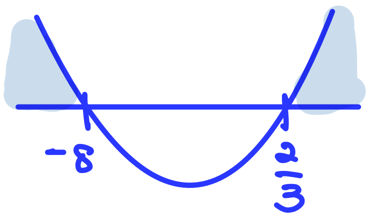
K1 SBS and expand

$$3x^2+22x-16 > 0$$ J1

$$(x+8)(3x-2) > 0$$

critical number,

$x = -8$, $x = \frac{2}{3}$



K1

$x \neq 5$, $\therefore (-\infty,-8) \cup (\frac{2}{3}, 5) \cup (5, \infty)$ J1

Total : 11m

③ a) $f(x) = 4^{x+2}$

$f(a) = 1$

$4^{a+2} = 1$

$4^{a+2} = 4^0$ K1 equate base

$a+2 = 0$

$a = -2$ J1

b) $f(x) = 4^{x+2}$

$ff^{-1}(x) = x$

$4^{f^{-1}(x)+2} = x$ K1

$(f^{-1}(x)+2)\ln 4 = \ln x$ K1 ln both sides

$$f^{-1}(x)+2 = \frac{\ln x}{\ln 4}$$

$$f^{-1}(x) = \frac{\ln x}{\ln 4} - 2$$ J1

$D_{f^{-1}(x)} = (0, \infty)$ B1

Total : 6m

4 a) $D_f = (-\infty, \infty)$ B1

$R_f = (3, \infty)$ K1 J1

b) $g[f(x)] = \frac{1}{2}\ln(e^{-2x}+2)$

$g(e^{-2x}+3) = \frac{1}{2}\ln(e^{-2x}+2)$ K1

let $u = e^{-2x}+3$

$x = -\frac{1}{2}\ln(u-3)$ K1 x as subject
J1 correct

$g(u) = \frac{1}{2}\ln(e^{-2(-\frac{1}{2}\ln(u-3))}+2)$

$= \frac{1}{2}\ln(e^{\ln(u-3)}+2)$

$g(u) = \frac{1}{2}\ln(u-3+2)$

$g(u) = \frac{1}{2}\ln(u-1)$ K1 simplify

$g(x) = \frac{1}{2}\ln(x-1)$ J1

c) $g(x) = \frac{1}{2}\ln(x-1)$

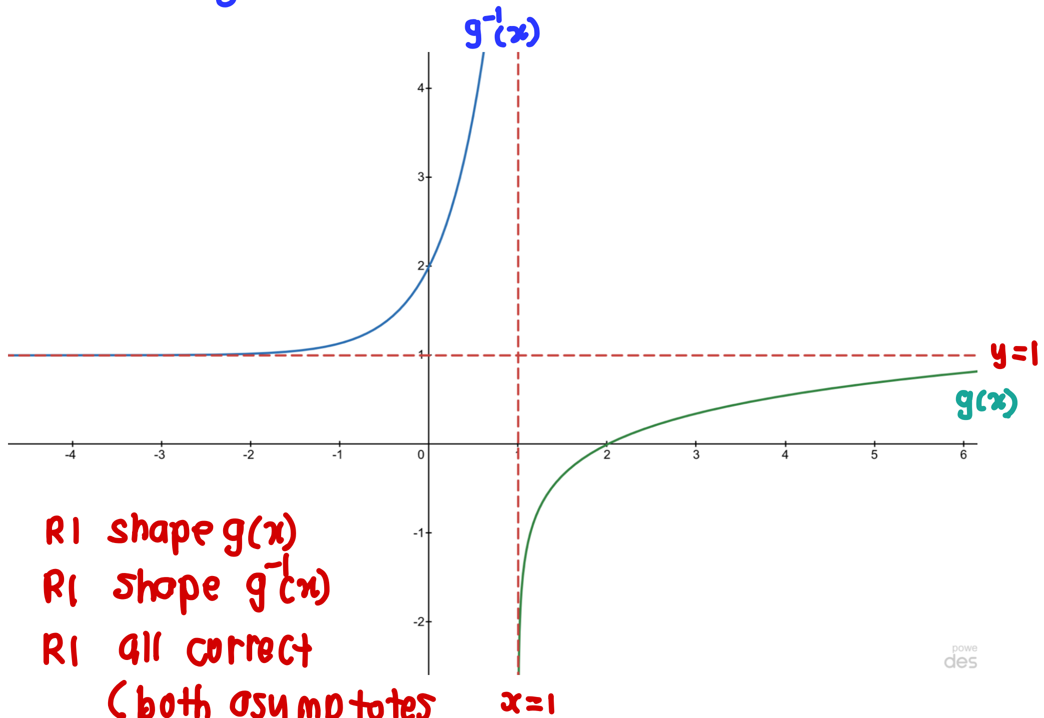
$g(g^{-1}(x)) = x$

$\frac{1}{2}\ln(g^{-1}(x)-1) = x$ K1

$\ln(g^{-1}(x)-1) = 2x$

$g^{-1}(x)-1 = e^{2x}$ K1

$g^{-1}(x) = e^{2x}+1$ J1



R1 shape $g(x)$
R1 shape $g^{-1}(x)$
R1 all correct
(both asymptotes correct)

Total : 14 m

⑤ a) $\lim_{x \to -\infty} \dfrac{3x+2}{\sqrt{4x^2-3}}$

$= \lim_{x \to -\infty} \dfrac{\dfrac{3x+2}{-x}}{\dfrac{\sqrt{4x^2-3}}{\sqrt{x^2}}}$ K1

$= \lim_{x \to -\infty} \dfrac{\dfrac{3x}{-x} + \dfrac{2}{-x}}{\sqrt{\dfrac{4x^2-3}{x^2}}}$

$= \lim_{x \to -\infty} \dfrac{-3 - \dfrac{2}{x}}{\sqrt{4 - \dfrac{3}{x^2}}}$ K1

$= \dfrac{-3-0}{\sqrt{4-0}}$ K1

$= -\dfrac{3}{2}$ J1

b) $f(5) = \lim_{x \to 5^+} f(x) = \lim_{x \to 5^-} f(x)$

$8 = \lim_{x \to 5^+} ax+3 = \lim_{x \to 5^-} x^2+bx-2a$

$\lim_{x \to 5^+} ax+3 = 8$ K1

$a(5)+3 = 8$

$5a = 5$

$a = 1$ J1

$\lim_{x \to 5^-} x^2+bx-2a = 8$ K1

$5^2+b(5)-2(1) = 8$

$b = -3$ J1

c) $y = \dfrac{x+1}{(x-4)(x+2)}$

Vertical asymptotes:

$(x-4)(x+2) = 0$

$x = 4, \quad x = -2$

$\lim_{x \to 4^+} \dfrac{x+1}{(x-4)(x+2)} = \dfrac{5}{0} = +\infty$

$\lim_{x \to 4^-} \dfrac{x+1}{(x-4)(x+2)} = \dfrac{5}{0} = -\infty$

K1 either one

$\lim_{x \to -2^+} \dfrac{x+1}{(x-4)(x+2)} = \dfrac{-1}{0} = +\infty$

$\lim_{x \to -2^-} \dfrac{x+1}{(x-4)(x+2)} = \dfrac{-1}{0} = -\infty$

K1 either one

Vertical asymptotes at $x=4$ and $x=-2$. J1

Horizontal asymptotes:

$\lim_{x \to +\infty} \dfrac{x+1}{x^2-2x-8}$

$= \lim_{x \to +\infty} \dfrac{\frac{x+1}{x^2}}{\frac{x^2-2x-8}{x^2}}$

$= \lim_{x \to +\infty} \dfrac{\frac{1}{x} + \frac{1}{x^2}}{1 - \frac{2}{x} - \frac{8}{x^2}}$

$= 0$

K1 working

$\lim_{x \to -\infty} \dfrac{x+1}{x^2-2x-8}$

$= \lim_{x \to -\infty} \dfrac{\frac{x+1}{x^2}}{\frac{x^2-2x-8}{x^2}}$

$= \lim_{x \to -\infty} \dfrac{\frac{1}{x} + \frac{1}{x^2}}{1 - \frac{2}{x} - \frac{8}{x^2}}$

$= 0$

K1 working

Horizontal asymptote at $y=0$ J1

Total : 14m

6) a) i.

$$y^3 = e^{x-y}$$

$$3y^2 \frac{dy}{dx} = e^{x-y}\left(1 - \frac{dy}{dx}\right)$$ K1 diff correctly both sides

$$3y^2 \frac{dy}{dx} = e^{x-y} - e^{x-y}\frac{dy}{dx}$$

$$3y^2 \frac{dy}{dx} + e^{x-y}\frac{dy}{dx} = e^{x-y}$$ K1

$$\frac{dy}{dx}\left(3y^2 + e^{x-y}\right) = e^{x-y}$$

$$\frac{dy}{dx} = \frac{e^{x-y}}{3y^2 + e^{x-y}}$$

$$\frac{dy}{dx} = \frac{y^3}{3y^2 + y^3}$$ K1

$$\frac{dy}{dx} = \frac{y^3}{y^2(3+y)}$$

$$\frac{dy}{dx} = \frac{y}{3+y}$$ J1 shown

ii.

$$\frac{dy}{dx} = \frac{y}{3+y}$$

$u = y$, $u' = \frac{dy}{dx}$

$v = 3+y$, $v' = \frac{dy}{dx}$

$$\frac{d^2y}{dx^2} = \frac{(3+y)\frac{dy}{dx} - y\frac{dy}{dx}}{(3+y)^2}$$ K1 J1

$$(3+y)^2 \frac{d^2y}{dx^2} = (3+y)\frac{dy}{dx} - y\frac{dy}{dx}$$

$$(3+y)^2 \frac{d^2y}{dx^2} = (3+y)\left(\frac{y}{3+y}\right) - y\left(\frac{y}{3+y}\right)$$ K1

$$(3+y)^2 \frac{d^2y}{dx^2} = \frac{y(3+y) - y^2}{3+y}$$

$$(3+y)^3 \frac{d^2y}{dx^2} = 3y + y^2 - y^2$$

$$(3+y)^3 \frac{d^2y}{dx^2} - 3y = 0$$ shown J1

b) $x = 2 + \sqrt{\cos 2t}$

$$\frac{dx}{dt} = \frac{1}{2}(\cos 2t)^{-1/2}(-2\sin 2t) \quad K1$$

$$\frac{dx}{dt} = -\frac{\sin 2t}{\sqrt{\cos 2t}}$$

$y = 1 - \sqrt{\sin 2t}$

$$\frac{dy}{dt} = -\frac{1}{2}(\sin 2t)^{-1/2}(2\cos 2t) \quad K1$$

$$\frac{dy}{dt} = -\frac{\cos 2t}{\sqrt{\sin 2t}}$$

$$\frac{dy}{dx} = -\frac{\cos 2t}{\sqrt{\sin 2t}} \times -\frac{\sqrt{\cos 2t}}{\sin 2t} \quad ✓ K1$$

$$= \left(\frac{\cos 2t}{\sin 2t}\right)^{3/2}$$

$$= (\cot 2t)^{3/2} \quad J1$$

$$\frac{d^2y}{dx^2} = \underbrace{\frac{3}{2}(\cot 2t)^{1/2}(-2\operatorname{cosec}^2 2t)}_{K1} \times -\frac{\sqrt{\cos 2t}}{\sin 2t} \quad K1$$

$$= \frac{3\sqrt{\cot 2t}\sqrt{\cos 2t}}{(\sin 2t)^3}$$

$$= \frac{3\sqrt{\frac{\cos 2t}{\sin 2t}}\sqrt{\cos 2t}}{(\sin 2t)^3} \quad K1 \text{ tukar kpd } \sin/\cos$$

$$= \frac{3\cos 2t}{(\sin 2t)^{7/2}} \quad J1$$

Total: 16m

7 a) $A = 4\pi r^2 + 2\pi r h$ K1

$= 4\pi r^2 + 2\pi r\left(\frac{90-4r^3}{3r^2}\right)$ K1 sub h

$= 4\pi r^2 + \frac{180\pi}{3r} - \frac{8r^2}{3}$

$= \frac{4\pi r^2}{3} + \frac{60\pi}{r}$ shown J1

b) $A = \frac{60\pi}{r} + \frac{4}{3}\pi r^2$

$\frac{dA}{dr} = -\frac{60\pi}{r^2} + \frac{8}{3}\pi r$ K1

$\frac{dA}{dr} = 0$

$-\frac{60\pi}{r^2} + \frac{8}{3}\pi r = 0$ K1

$\frac{8}{3}\pi r = \frac{60\pi}{r^2}$

$r^3 = \frac{45}{2}$

$r = 2.82$ J1

$\frac{d^2A}{dr^2} = \frac{120\pi}{r^3} + \frac{8}{3}\pi$ K1

when $r = 2.82$,

$\frac{d^2A}{dr^2} = \frac{120\pi}{(2.82)^3} + \frac{8}{3}\pi$ K1

$\frac{d^2A}{dr^2} = 25.133 > 0$

$\therefore$ $r = 2.82$ will give minimum surface area. J1

Total : 9m